Preliminary Summary of Acceleration-Basis and Localization Exchange in Exchange-Interference Scattering-Matrix Fermion-Like Collisions v1

Subtitle: Localization redistribution in one-sided high-harmonic recursive scattering while preserving acceleration-like readout
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Abstract

This paper summarizes preliminary experiments in which the exchange-interference scattering-matrix version of the fermion-like recoil map established in the V2 series is placed on the basis of the AB two-body acceleration-like readout.

The experiments cover the low odd-harmonic bottom, readout-stop control, removal of name hair, one-sided high-harmonic conditions, recursive scattering, and reflection-rate sweeps.

The acceleration V2 basis was preserved. The odd-harmonic bottom was `N=1`, both with and without name hair. In the readout-stop control, the difference in the localization index `L` was `0.0`.

When the one-sided high-harmonic condition `N_A=1`, `N_B=63` was inserted into recursive scattering, the difference between low-order and high-order waveforms was preserved at the complete-transmission endpoint `R=0` and the complete-reflection endpoint `R=1`. In contrast, for intermediate scattering `0<R<1`, the localization index `L` and the effective harmonic order `N_eff` were exchanged periodically.

In the reflection-rate sweep, `R=0.70` gave the minimum `L` difference, `2.48676e-05`, with `N_eff` difference `0.153386`. This is recorded as a recursive point at which the localization indices and effective harmonic orders of the two channels become close under the one-sided high-harmonic condition.

Within the range of this experiment, stationary convergence was not observed. The observed behavior was periodic localization exchange generated by a two-channel scattering matrix without a dissipative term.

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1. Purpose of the Experiment

The purpose of this experiment is the following:

```
To read whether a one-sided high harmonic moves into the localization readout
of the other channel on a fermion-like recoil map that preserves acceleration-like readout.
```

Here, localization is not the position of a particle placed in an external space. It is a quantity read from the `chi`-direction density, harmonic distribution, effective harmonic order, and localization index constructed inside the closed phase system.

The readout targets are the following quantities on the scattering-matrix outgoing channels:

```
L
N_eff
rho_chi
expected_origin_A
expected_origin_B
```

The experiment records how these quantities are redistributed on the outgoing channels of the scattering matrix.

---

2. Basis Map

From the exchange-interference phase `Delta_F`, define the transmission amplitude `t` and reflection amplitude `r`:

```
t_\Delta
=
e^{i\Delta_F/2}
\cos\frac{\Delta_F}{2}
```

```
r_\Delta
=
-i e^{i\Delta_F/2}
\sin\frac{\Delta_F}{2}
```

The reflection and transmission rates are read as

```
R=|r_\Delta|^2
```

```
T=|t_\Delta|^2.
```

The two-channel output in this experiment is

```
minus_out = r A_ref + t B_trans
plus_out  = t A_trans + r B_ref
```

In the recursive experiment, the outgoing channels are passed to the next input:

```
A_{k+1}
=
\operatorname{norm}
\left(
r A_k + t B_k
\right)
```

```
B_{k+1}
=
\operatorname{norm}
\left(
t A_k + r B_k
\right).
```

This recursive map contains no dissipative term.

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3. Acceleration V2 Basis

This experiment uses the AB two-body acceleration V2 result as its basis.

Quantity	Value
fermionic_reflection_rate	1.0
fermionic_transmission_rate	3.749399456654644e-33
fermionic_q_out_factor	-1.0
max_Q_closed_abs	0.0

Quantity	Value
label_free_pass_vs_fermionic_match_all_cases	`true`
fermionic_regular_cell_harmonic_consistent_nonstrong_modes	`true`
fermionic_c1_area_sweep_detected_all_cases	`true`

Therefore, the scattering-matrix version of the fermion-like recoil map preserves the basis of the acceleration-like readout.

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4. Low Odd-Harmonic Bottom

Under complete reflection `Delta_F=pi`, the odd-harmonic order `N` was lowered.

Condition	Bottom
With name hair	`N=1`
Without name hair	`N=1`

Even at `N=1`, the direction readout and internal readout did not break down.

Representative values are as follows:

N	hair	p_chi	d_q	chi center-cell mass	chi peak contrast	L	total name hair
1	true	`-1`	`4.44089e-16`	`0.5`	`2`	`0.000183105`	`1`
1	false	`-1`	`8.88178e-16`	`0.5`	`2`	`0.000183105`	`0`

From this result, the readout bottom under the present experimental conditions is read as `N=1`.

Under the present experimental conditions, increasing the odd-harmonic order mainly appeared as a sharpening of the `chi`-direction density, rather than as the presence or absence of information.

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5. One-Sided High-Harmonic Condition

Using the bottom `N_A=1` as the reference, the experiment examined a condition in which only one side has high harmonics.

The representative condition is

```
N_A = 1
N_B = 63
```

In one-step scattering with `R=0.55`, `T=0.45`, the outgoing channels retained the allocation of A-origin and B-origin components.

channel	R	T	N_eff	L	origin_A	origin_B
minus_out	`0.55`	`0.45`	`14.95`	`0.00135002`	`0.55`	`0.45`
plus_out	`0.55`	`0.45`	`18.05`	`0.0018526`	`0.45`	`0.55`

At the one-step scattering stage, the one-sided high harmonic is allocated to the outgoing channels.

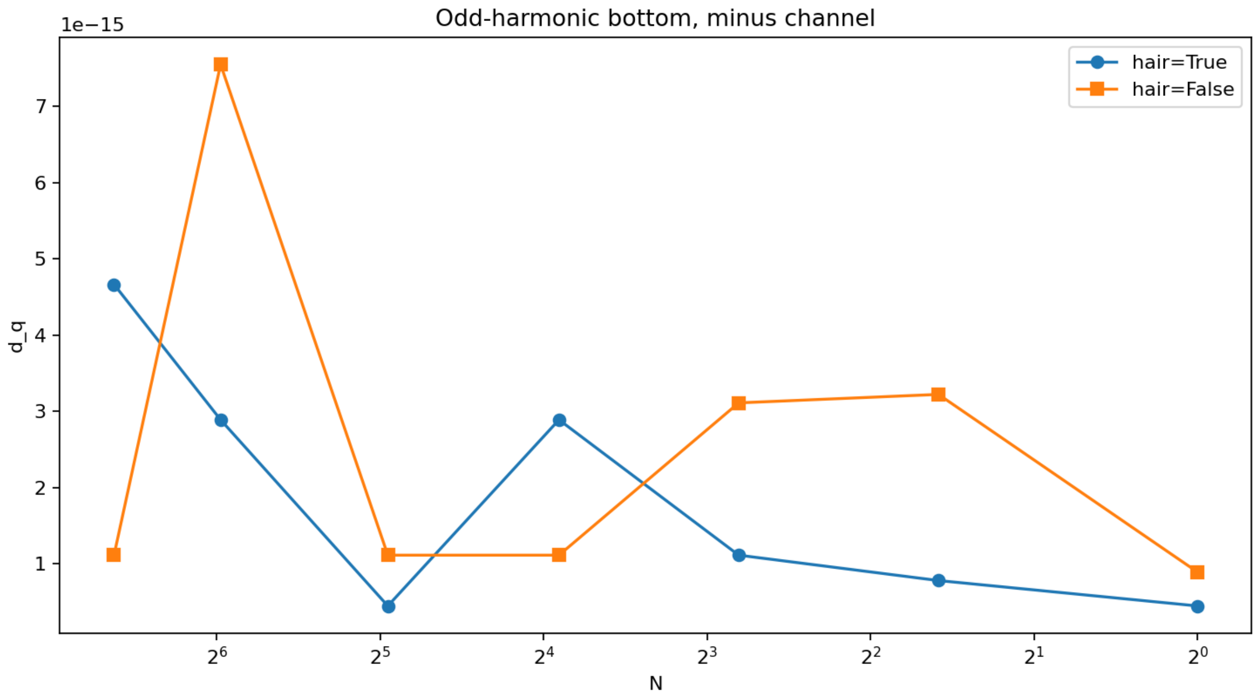

Figure 1: low N

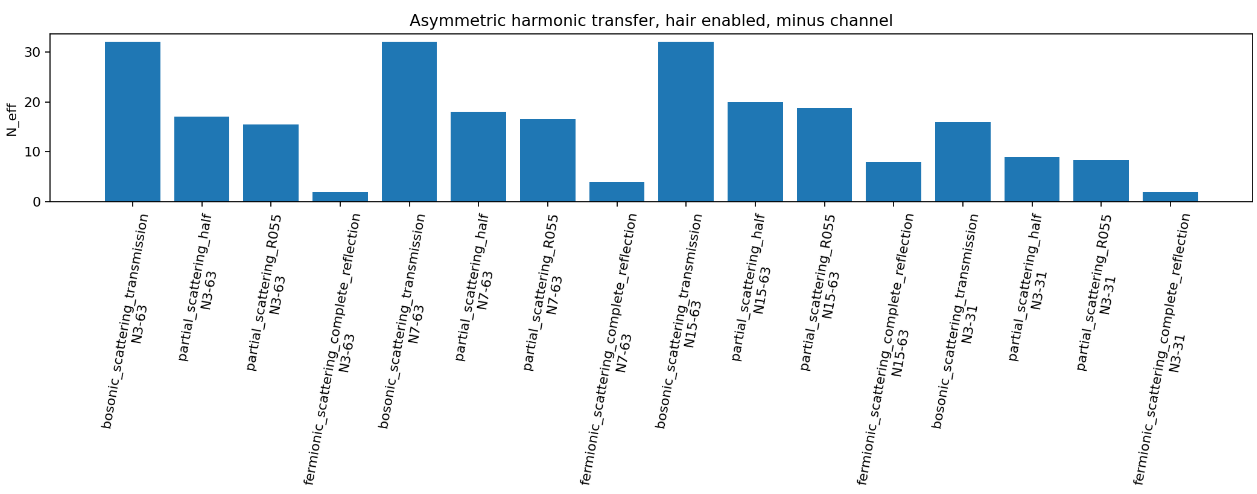

Figure 2: one side high harmonic

6. Recursive One-Sided High Harmonic

The outgoing channels were passed to the next input, and recursive scattering was performed up to 128 collisions.

The representative result for `R=0.55`, `T=0.45` is as follows:

collision	A: L	A: N_eff	B: L	B: N_eff
0	0.000183105	1	0.00520897	32
16	0.00167148	16.9941	0.00151127	16.0059
32	0.00519937	31.9685	0.000183722	1.0315
64	0.000185626	1.12587	0.00517067	31.8741
128	0.000194012	1.50145	0.00505728	31.4985

The localization index and effective harmonic order did not converge to a stationary value.

The recursive map periodically exchanged the low-order side and the high-order side.

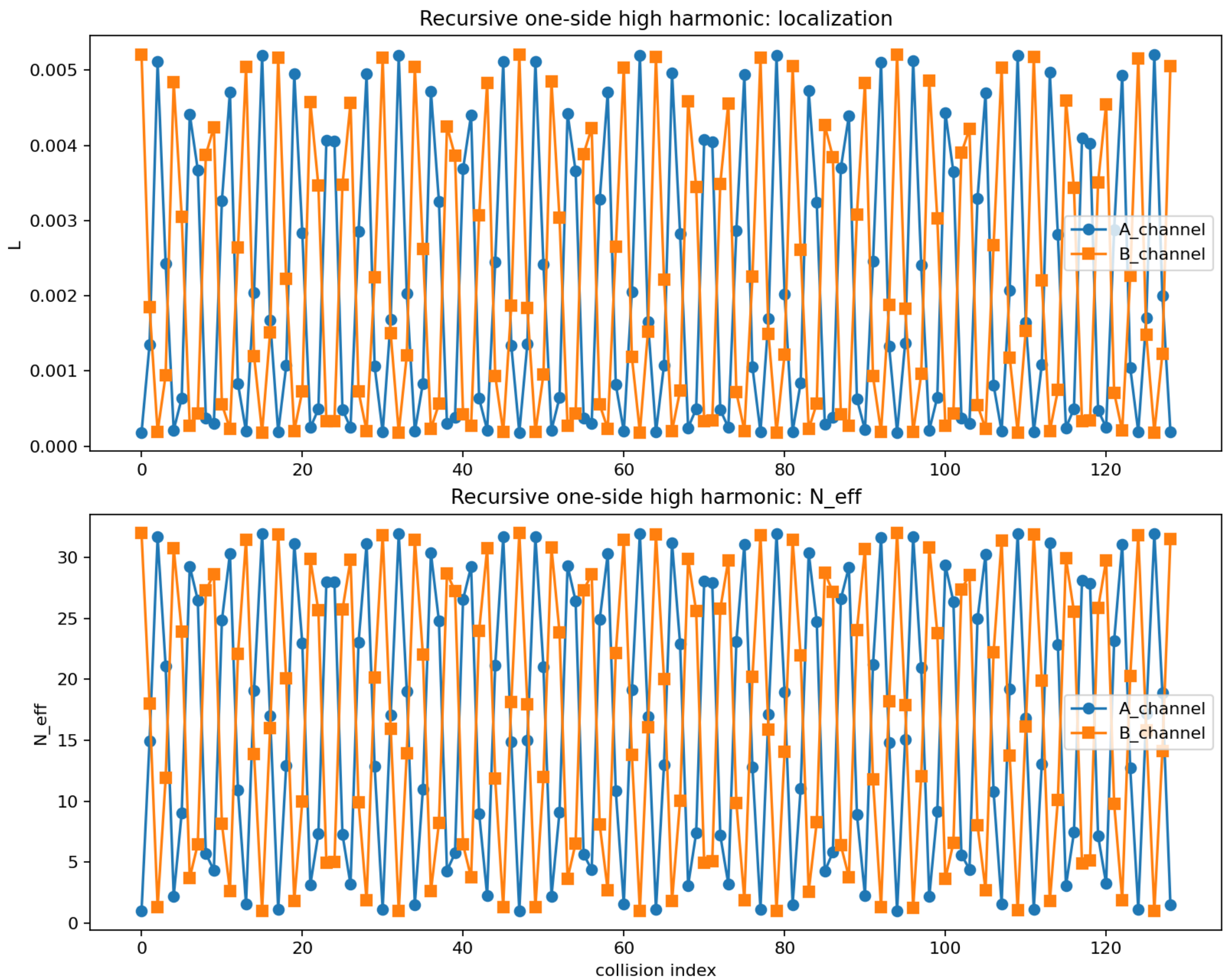

Figure 3: recursive localization transfer

7. Reflection-Rate Sweep

The reflection rate `R` was swept under the condition `N_A=1`, `N_B=63`.

R	T	minimum L difference	collision at minimum L difference	N_eff difference at minimum L difference
0	1	0.00502586	1	31
0.51	0.49	5.37371e-05	78	0.331455
0.55	0.45	0.000114791	110	0.708042
0.6	0.4	0.000185909	125	1.1467
0.7	0.3	2.48676e-05	42	0.153386
0.9	0.1	8.21216e-05	61	0.506534
1	3.7494e-33	0.00502586	126	31

At the complete-transmission endpoint `R=0`, the difference between the low-order waveform and the high-order waveform was preserved.

At the complete-reflection endpoint `R=1`, the difference between the low-order waveform and the high-order waveform was also preserved.

For intermediate scattering `0<R<1`, recursive points appeared at which the differences in `L` and `N_eff` became small.

In this sweep, `R=0.70` gave the minimum `L` difference.

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8. Waveform Localization Snapshots

The reduced density in the `chi` direction is read as

```
\rho_\chi
=
\sum_\eta |\psi(\chi,\eta)|^2.
```

In the figures, each line is amplitude-normalized so that its maximum value is `1`.

The figure compares the complete-transmission endpoint `R=0`, the intermediate scattering case `R=0.70`, and the complete-reflection endpoint `R=1`.

At `R=0`, A and B are exchanged on the fixed channels. However, the difference between the low-order waveform and the high-order waveform itself is preserved.

At `R=1`, the difference between the low-order waveform and the high-order waveform is preserved even on the fixed channels.

At `R=0.70`, the localization indices of the two channels become close at collision `42`.

Condition	collision	A: L	A: N_eff	B: L	B: N_eff
R=0	0	0.000183105	1	0.00520897	32
R=0	1	0.00520897	32	0.000183105	1
R=0.70	0	0.000183105	1	0.00520897	32
R=0.70	42	0.00160271277972861.352866927584705640.00157784520896556.0462330724152941			
R=1	0	0.000183105	1	0.00520897	32
R=1	1	0.000183105	1	0.00520897	32

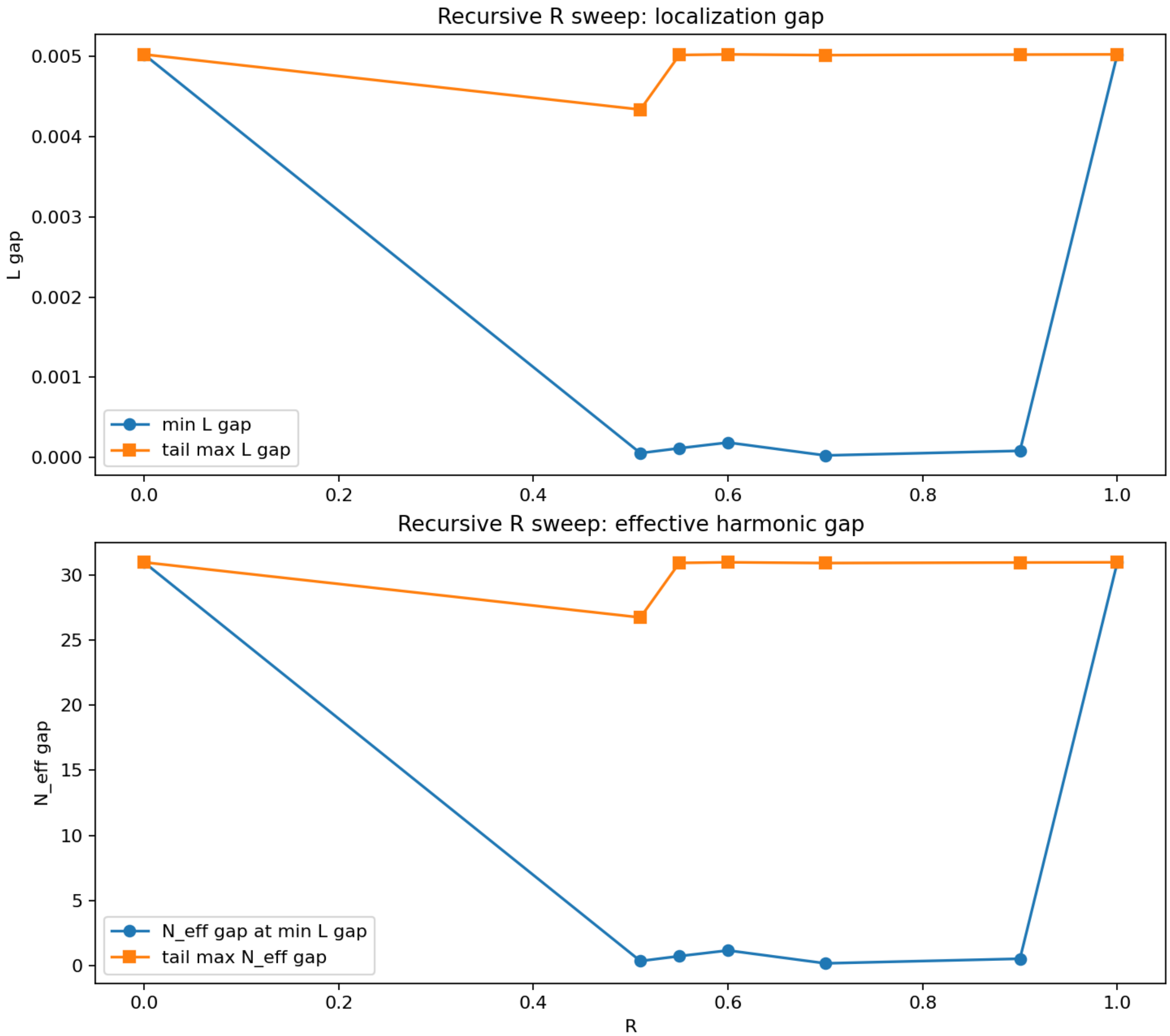

Figure 4: recursive R sweep

This figure records, as amplitude-normalized density waveforms, that localization redistribution appears in intermediate scattering and not at the endpoints.

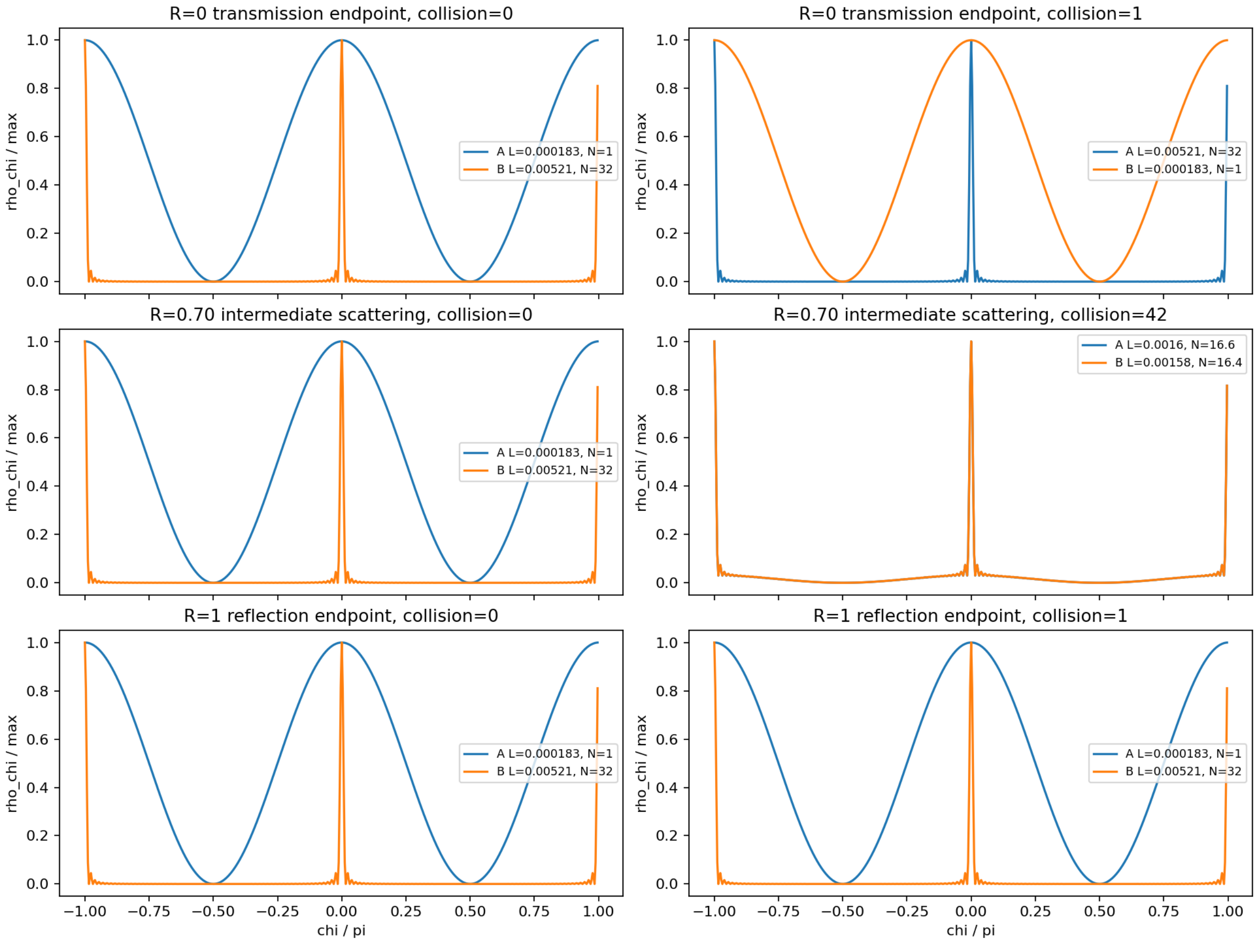

Figure 5: waveform localization snapshots

8.1 Localization-Exchange Process at R=0.70

For `R=0.70`, the figure shows `rho_chi / max` at collision counts `0,1,2,3,5,10,20,42`.

At `collision=42`, `L` and `N_eff` for A/B are close, and the localization indices of the two waveforms become close.

The process read from the figure is not monotonic localization, but an exchange oscillation of localization.

collision	State read from the figure
0	A starts as a low-order base waveform, while B starts as a high-order localized waveform.
1	A receives high-order components, and the waveform peak of A sharpens.
2,3	A moves toward the high-localization side, while B moves toward a dull shape close to the low-order base waveform.
5,10	A becomes dull again, and the high-localization shape returns to the B side.

collision	State read from the figure
42	The effective orders of A/B become close, `16.6` and `16.4`, and their localization indices become close.

From this figure, the one-sided high-localization shape is not simply averaged away in one step. Instead, it is read as moving back and forth between A and B through the recursive action of intermediate scattering. Near `collision=42`, the effective order is lower than the initial B value `N_eff=32`, while the amplitude-normalized waveform still retains a sharp shape that is not only the low-order base wave.

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9. Readout-Stop Control

In the readout-on/readout-off control,

`observation_stop_max_L_delta = 0.0`

was obtained.

Under the readout-on and readout-off conditions, the localization index `L` was recorded with the same value.

The recursive map in this experiment contains no dissipative term depending on readout-wave strength.

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10. Conclusion

The results of this experiment are as follows:

```
1. The acceleration V2 basis was preserved.
2. The odd-harmonic bottom was N=1.
3. Removing name hair did not change the N=1 bottom.
4. At complete transmission R=0, the localization difference was preserved.
5. At complete reflection R=1, the localization difference was also preserved.
6. At intermediate scattering 0<R<1, the localization index L and effective harmonic order N_eff were re
7. At R=0.70, L and N_eff of A/B became close at collision 42.
8. Within the range of this experiment, periodic exchange, not stationary convergence, was observed.
```

Therefore, localization redistribution in the fermion-like recoil map appears not at the complete-transmission or complete-reflection endpoints, but in intermediate scattering with both reflection and transmission components.

This provides an experimental basis for examining changes in waveform localization from the internal outgoing channels of the exchange-interference scattering matrix.

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11. Outputs

Type	File
Script	`run_exchange_scattering_matrix_fermionic_localization_tra`
JSON	`exchange_scattering_matrix_fermionic_localization_transfe`
CSV	`exchange_scattering_matrix_fermionic_localization_transfe`
Validation memo	`exchange_scattering_matrix_fermionic_localization_exchang`
Low odd-harmonic bottom figure	`exchange_scattering_matrix_fermionic_localization_transfe`
One-sided high-harmonic figure	`exchange_scattering_matrix_fermionic_localization_transfe`

Type	File
Waveform localization snapshots	exchange_scattering_matrix_fermionic_localization_transfer
R=0.70 waveform evolution	exchange_scattering_matrix_fermionic_localization_transfer
Recursive R sweep figure	exchange_scattering_matrix_fermionic_localization_transfer
Recursive localization figure	exchange_scattering_matrix_fermionic_localization_transfer

References

Self-Citations

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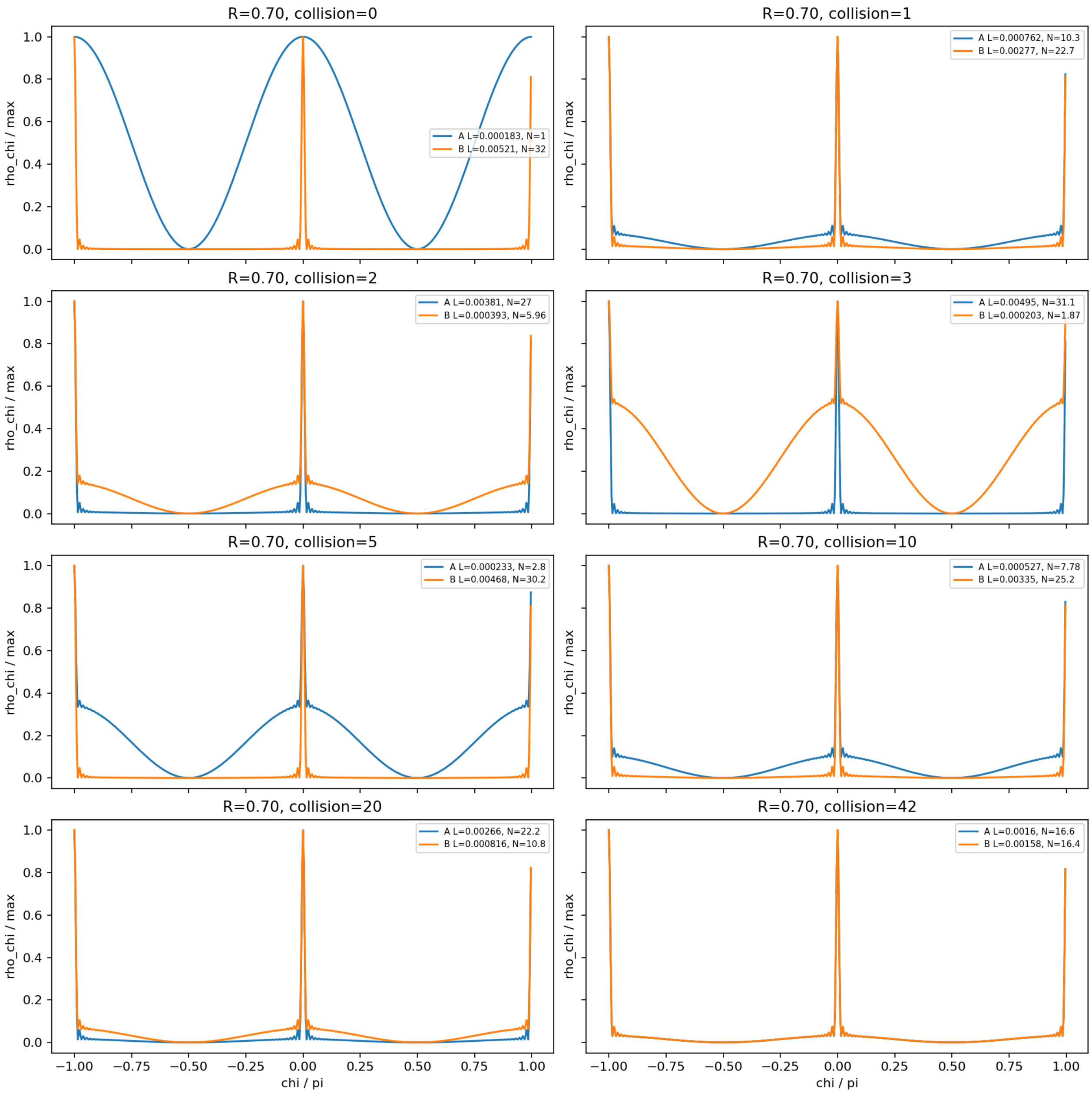

Figure 6: R070 waveform evolution